

NOMAD : New Operations & Market Advisory Director

On Device AI Consultant, Planner & Operational Partner for the Physical World

1. Concept Overview

What Is It?

An on-device AI startup advisor, consultant, planner, and operational partner that helps businesses make better decisions about their physical locations, facilities, operations, logistics, risks, and expansion.

The Core Idea

The product uses Mireye as its physical-world intelligence layer, allowing the advisor to understand and reason about real-world locations, surrounding demographics, zoning, permits, hazards, accessibility, proximity, and other relevant physical attributes.

Not Just a Dashboard

Rather than being a collection of separate tools, the product acts as a continuous business partner that can analyze a company's locations and answer questions such as:

1. How should our physical network be optimized?
2. What regulations apply?
3. What risks exist around our locations?
4. Is this facility suitable?
5. Where should we expand?

2. Product Positioning

Role	What It Means	Example Questions It Answers
AI Consultant	Provides expert guidance on physical-world decisions	"Which site best fits our criteria?"

Planner	Creates actionable plans for expansion, logistics, returns	“How should we structure our returns network?”
Analyst	Continuously monitors and compares location contexts	“What risks changed at our facilities this quarter?”
Operational Partner	Integrates with business systems for ongoing decision support	“Which locations need local exceptions to our standard policy?”

A client could give it their locations, facilities, suppliers, warehouses, and expansion plans, and continuously ask it to analyze, compare, verify, monitor, and recommend.

3. Geographic Scope

The initial product is matching Mireye’s current geographic focus.

1. Global expansion is a future consideration, not an initial requirement
2. External data sources used for market-estimation capabilities (*) and networking/real-time capabilities (NT) should primarily complement the USA-focused Mireye data
3. Primary: All 50 US states + territories

Capabilities marked with (*) rely on external free/public data for demand estimation.

Capabilities marked with (NT) require networking/real-time external feeds. Both should focus on USA-relevant data sources initially.

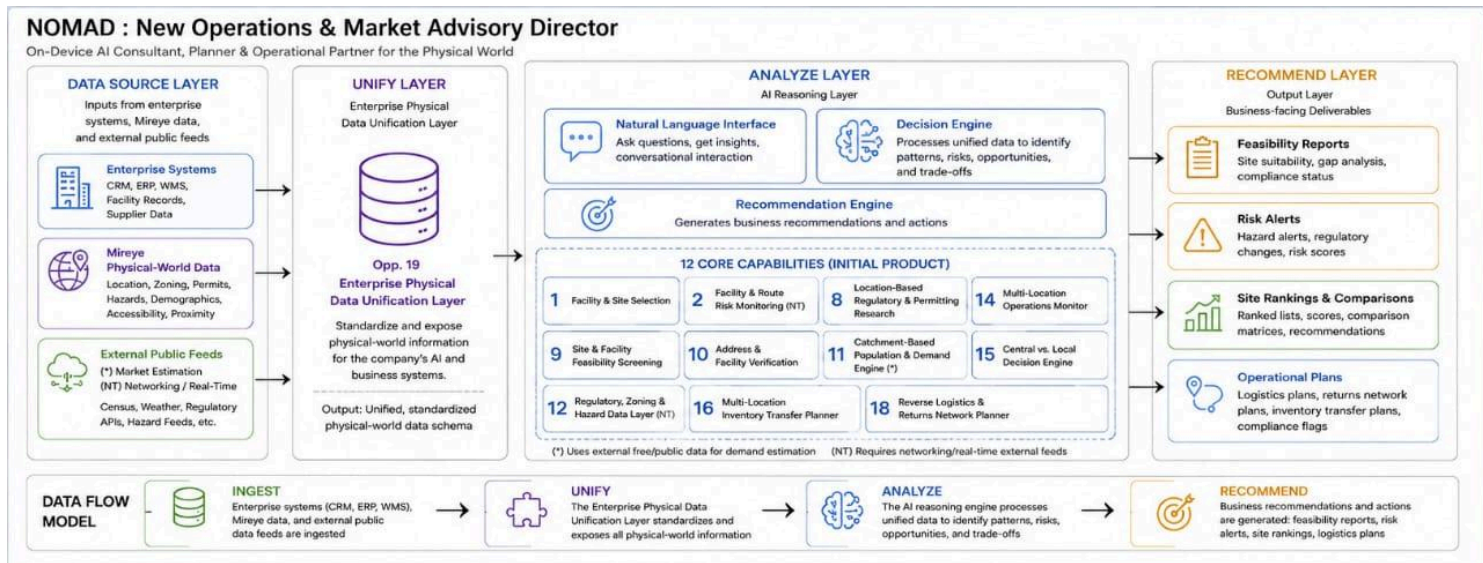
4. Core Differentiator

The central advantage is that Mireye is not merely an optional data source. It forms the physical world intelligence foundation underneath the advisor.

Enterprise Data + Mireye + External Supporting Data → AI Reasoning → Business Recommendation / Action

This layered approach means Mireye is embedded in every capability - not bolted on as a feature. Every recommendation, ranking, risk alert, and feasibility assessment flows through Mireye’s physical-world understanding.

5. System Architecture



Architecture Layers

Layer	Description	Key Components
AI Reasoning Layer	Top-level intelligence that processes unified data and generates business recommendations	Natural language interface, decision engine, recommendation engine
Mireye Intelligence Layer	Physical-world intelligence foundation - standardizes and exposes physical-world information	Location data, zoning, permits, hazards, demographics, accessibility
Data Source Layer	Inputs from enterprise systems, Mireye data, and external public feeds	Enterprise (CRM, ERP, WMS), Mireye, External APIs (census, weather, real-time feeds)
Output Layer	Business-facing deliverables	Feasibility reports, risk alerts, site rankings, logistics plans, compliance flags

6. Data Flow Model

Flow Stages

1. Recommend - Business recommendations and actions are generated: feasibility reports, risk alerts, site rankings, logistics plans
2. Analyze - The AI reasoning engine processes unified data to identify patterns, risks, opportunities, and trade-offs
3. Unify - The Enterprise Physical Data Unification Layer standardizes and exposes all physical-world information consistently
4. Ingest - Enterprise systems (CRM, ERP, WMS), Mireye physical-world data, and external public data feeds are ingested

7. The 12 Core Capabilities (Initial Product)

The first version combines the following 12 capabilities. Each is mapped to its opportunity reference number.

Enterprise Physical Data Unification Layer

Type: Core Foundation

Description: “Standardize and expose physical-world information for the company’s AI and business systems. This is the foundational layer that makes all other capabilities possible.”

Input	Output
Enterprise location data, facility records, supplier addresses	Unified, standardized physical-world data schema

Site & Facility Feasibility Screening

Type: Core

Description: “Determine whether candidate facilities/sites satisfy physical and geographic requirements. Screens locations against criteria such as accessibility, zoning, infrastructure, and environmental factors.”

Input	Output
Candidate site addresses, business requirements	Feasibility scores, pass/fail indicators, gap analysis

Regulatory, Zoning & Hazard Data Layer (NT)

Type: NT (Networking/Real-Time)

Description: “Monitor zoning, permits, hazards, and other location-level risks. Enhanced with live external feeds for real-time regulatory and hazard updates.”

Input	Output
Location coordinates, regulatory feeds	Regulatory status, hazard alerts, compliance flags

Location-Based Regulatory & Permitting Research

Type: Core

Description: “Investigate location-specific zoning and permitting requirements. Provides detailed research on what permits are needed, what zoning applies, and what restrictions exist for a given address.”

Input	Output
Address, business type, intended use	Required permits, zoning classification, restrictions, application guidance

Address & Facility Verification

Type: Core

Description: “Verify that addresses correspond to legitimate, appropriately classified physical locations. Prevents errors from invalid, misclassified, or non-existent addresses in enterprise systems.”

Input	Output
Address strings from enterprise systems	Verification status, corrected addresses, facility classification

Facility & Route Risk Monitoring (NT)

Type: NT (Networking/Real-Time)

Description: “Identify geographic risks affecting facilities, suppliers, and routes. Enhanced with

live external feeds for weather, natural disasters, and other real-time risk factors.”

Input	Output
Facility locations, supplier locations, route paths	Risk scores, active alerts, recommended mitigations

Multi Location Operations Monitor

Type: Core

Description: “Compare and monitor the physical/regulatory context of an entire location network. Provides a dashboard view across all company locations for comparative analysis.”

Input	Output
All company location data	Comparative metrics, outlier alerts, network-level insights

Facility & Site Selection

Type: Core

Description: “Rank potential locations based on physical, geographic, regulatory, and demographic suitability. The flagship decision-support capability for expansion planning.”

Input	Output
Candidate locations, weighted criteria	Ranked list with scores, comparison matrix, recommendation

Central vs. Local Decision Engine

Type: Core

Description: “Identify where standardized company decisions may require local exceptions because of geographic differences. Helps balance corporate standardization with local compliance and optimization.”

Input	Output
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Corporate policies, location-specific data	Exception flags, local adaptation recommendations
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Catchment-Based Population & Demand Engine (*)

Type: * (External Public Data Dependent)

Description: “Analyze surrounding populations and use external free/public data to estimate geographic demand. Combines Mireye demographic data with census and other public sources.”

Input	Output
Location, catchment radius	Population demographics, demand estimates, market potential

Reverse Logistics & Returns Network Planner

Type: Core

Description: “Determine geographically suitable destinations for returns, repairs, refurbishment, recycling, and disposal. Optimizes the reverse supply chain based on physical-world factors.”

Input	Output
Return origins, processing facility locations	Optimal routing, facility assignments, cost/time estimates

Multi Location Inventory Transfer Planner

Type: Core

Description: “Use geographic relationships and accessibility to optimize potential inventory transfers between locations. Considers distance, accessibility, regulatory constraints, and risk.”

Input	Output
Inventory levels by location, transfer constraints	Transfer recommendations, optimal routes, priority rankings

8. Future Product Additions

Once the core platform is established, additional capabilities can be added as modules. These should be treated as future extensions, rather than core requirements for the initial product.

Module	Description	Priority
Supplier Geographic Concentration Mapping	Visualize and analyze supplier geographic clustering risks	High
Location-Aware Workforce Planner	Plan staffing based on location-specific labor factors	Medium
Retail/Distribution Site & Competitor Proximity Mapping	Map competitor proximity for retail and distribution decisions	High
Local Demand / Market Sizing (*)	Deep market sizing using external demographic and economic data	High
Regional Customer Service Context Engine	Provide location-aware context for customer service operations	Medium
Manufacturer/Facility Verification	Extended verification for manufacturing facilities and certifications	Medium
Labor Market Sizing by Location (*)	Analyze labor availability, costs, and skills by geography	Medium

9. Product Roadmap

Phase 1: Core Platform (Initial Release)

Deliver all 12 core capabilities with the Mireye intelligence layer as the foundation.

Phase 2: Module Extensions

Add supplier concentration mapping, competitor proximity, and workforce planning modules based on client demand.

Phase 3: Advanced Analytics & Market Sizing

Expand into deep market sizing, labor market analysis, and regional customer service context capabilities.

10. Two-Week Action Plan

The following plan outlines development priorities for the next 14 days, from **August 15 through August 28, 2026**.

Team Roles

Member	Primary Responsibility
Vedant Awasthi : Backend & Mireye Integration	Backend architecture, data ingestion, data normalization, Mireye API integration
Shaurya Suyal : AI & Decision Engine	AI reasoning, scoring, ranking, recommendation logic, capability-specific decision models
Paarth Gupta : Frontend & Product	Dashboard, visualization, user workflow, demo experience, documentation/presentation

Week 1 : Foundation & Core Integration

Days 1-2 (Aug 15-16): Foundation & Architecture

#	Task	Owner	Status
1	Define overall technical architecture and module boundaries	Vedant Awasthi : Backend & Mireye Integration	Not Started
2	Set up repository structure and development environment	Vedant Awasthi : Backend & Mireye Integration	Not Started
3	Define unified physical-world data schema	Vedant Awasthi : Backend & Mireye Integration	Not Started
4	Define Mireye API integration points and required endpoints	Vedant Awasthi : Backend & Mireye Integration	Not Started
5	Define AI reasoning workflow and recommendation pipeline	Shaurya Suyal : AI & Decision Engine	Not Started
6	Define scoring/ranking methodology for site and location decisions	Shaurya Suyal : AI & Decision Engine	Not Started
7	Design initial user workflow and dashboard structure	Paarth Gupta : Frontend & Product	Not Started
8	Create project board, task breakdown, and sprint tracking	Paarth Gupta : Frontend & Product	Not Started

Days 3-4 (Aug 17-18): Enterprise Physical Data Layer

Vedant Awasthi : Backend & Mireye Integration

#	Task	Status
9	Build core enterprise-data ingestion pipeline	Not Started
10	Implement data standardization and normalization	Not Started
11	Create unified physical-world data schema and validation	Not

	rules	Started
12	Build initial CSV/JSON input connectors	Not Started

Shaurya Suyal : AI & Decision Engine

#	Task	Status
13	Define how unified location data will be consumed by the AI layer	Not Started
14	Design location-context representation for AI reasoning	Not Started
15	Define recommendation/scoring interface between backend and AI layer	Not Started
16	Prepare initial reasoning prompts for location analysis	Not Started

Paarth Gupta : Frontend & Product

#	Task	Status
17	Build initial dashboard shell	Not Started
18	Create location input/import interface	Not Started
19	Design location-profile display	Not Started
20	Design basic recommendation/results interface	Not Started

Day 5 (Aug 19): Mireye Integration

Vedant Awasthi : Backend & Mireye Integration

#	Task	Status
21	Implement Mireye API client and authentication	Not

		Started
22	Build Mireye → unified-schema data mapper	Not Started
23	Test Mireye retrieval for sample USA addresses	Not Started
24	Document available Mireye fields, coverage, and known gaps	Not Started

Shaurya Suyal : AI & Decision Engine

#	Task	Status
25	Connect AI reasoning interface to standardized location data	Not Started
26	Test basic location-context reasoning using Mireye-derived data	Not Started

Paarth Gupta : Frontend & Product

#	Task	Status
27	Connect dashboard location input to backend	Not Started
28	Display Mireye-derived location information in the UI	Not Started

Team

#	Task	Status
29	Week 1 integration check and module handoff verification	Not Started

Week 2 : Capability Build & Validation

Days 6-8 (Aug 22-24): Core Capability Build

Vedant Awasthi : Backend & Mireye Integration

#	Task	Status
30	Implement Address & Facility Verification backend	Not Started
31	Implement Site & Facility Feasibility data pipeline	Not Started
32	Build Regulatory & Zoning data retrieval scaffold	Not Started
33	Implement required backend endpoints for capability modules	Not Started

Shaurya Suyal : AI & Decision Engine

#	Task	Status
34	Build basic Site Selection scoring/ranking engine	Not Started
35	Build feasibility scoring logic	Not Started
36	Build regulatory/hazard interpretation logic	Not Started
37	Implement initial AI recommendation interface	Not Started

Paarth Gupta : Frontend & Product

#	Task	Status
38	Build Address & Facility Verification UI	Not Started
39	Build Site & Facility Feasibility UI	Not Started

40	Build Regulatory/Zoning results interface ²	Not Started
41	Build Multi-Location Operations Monitor dashboard scaffold	Not Started

Days 9-10 (Aug 25-26): Testing & Validation

Vedant Awasthi : Backend & Mireye Integration

#	Task	Status
42	End-to-end test of data ingestion and Mireye pipeline	Not Started
43	Test Mireye integration using representative USA addresses	Not Started
44	Validate API error handling and missing-data behavior	Not Started

Shaurya Suyal : AI & Decision Engine

#	Task	Status
45	Validate feasibility scoring against known good/bad sites	Not Started
46	Test site-selection ranking consistency	Not Started
47	Evaluate AI recommendations for obvious hallucinations/inconsistencies	Not Started

Paarth Gupta : Frontend & Product

#	Task	Status
48	Test complete user workflow from input → result	Not Started
49	Validate dashboard usability and information hierarchy	Not Started

50	Fix frontend issues and improve demo workflow	Not Started
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Days 11-12 (Aug 27): Documentation & Demo Preparation

Vedant Awasthi : Backend & Mireye Integration

#	Task	Status
51	Document backend architecture and Mireye integration	Not Started
52	Document data flow and API interfaces	Not Started
53	Prepare representative USA demo dataset	Not Started

Shaurya Suyal : AI & Decision Engine

#	Task	Status
54	Document scoring, ranking, and reasoning methodology	Not Started
55	Prepare example decision scenarios and expected outputs	Not Started
56	Validate final AI recommendation flow for demo	Not Started

Paarth Gupta : Frontend & Product

#	Task	Status
57	Polish dashboard and demo interface	Not

		Started
58	Build complete demo flow: ingest → unify → analyze → recommend	Not Started
59	Prepare stakeholder presentation/deck	Not Started

Days 13-14 (Aug 28):Review

Vedant Awasthi : Backend & Mireye Integration

#	Task	Status
60	Verify backend stability for final demo	Not Started
61	Resolve integration/API issues discovered during rehearsal	Not Started
62	Update architecture documentation with lessons learned	Not Started

Shaurya Suyal : AI & Decision Engine

#	Task	Status
63	Conduct final validation of recommendations	Not Started
64	Tune scoring/reasoning based on demo results	Not Started
65	Define next AI capabilities for Sprint 3	Not Started

Paarth Gupta : Frontend & Product

#	Task	Status
66	Conduct final demo rehearsal	Not Started

67	Present working prototype to stakeholders	Not Started
68	Collect and organize stakeholder feedback	Not Started

Team

#	Task	Owner	Status
69	Prioritize next-sprint improvements	All Members	Not Started
70	Define Sprint 3 goals and ownership	All Members	Not Started

11. Key Assumptions & Risks

Key Assumptions

1. USA-focused scope is sufficient for initial client adoption
2. Enterprise systems can export location/facility data in standardized or adaptable formats
3. External public data sources (census, weather feeds, regulatory APIs) are accessible and sufficiently current
4. On-device AI processing can handle the reasoning complexity required for business recommendations
5. Mireye provides adequate USA coverage for zoning, permits, hazards, and demographic data

Key Risks

Risk	Impact	Likelihood	Mitigation
Mireye data coverage gaps in certain regions/states	High	Medium	Identify gaps early; supplement with external sources
External data source API changes or rate limits	Medium	Medium	Build adapter pattern for easy source swapping; cache

AI reasoning quality insufficient for complex decisions	High	Low	Start with structured decision trees; add AI incrementally
On-device performance constraints	Medium	Medium	Optimize data loading; consider hybrid approach
Enterprise data quality issues	Medium	High	Build robust data cleaning; flag issues
Scope creep	Medium	High	Strict backlog management; defer modules